

Benchmarking Language Models for Scale-C

Model Selection for H5P-Based Computer Security Education

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Abstract

Scale-C is an educational project that teaches computer security through interactive H5P units. The platform may eventually rely on a large language model to author content, answer learner questions, and support German-language material. Before adopting a specific model, it is necessary to establish whether a cybersecurity fine-tuned variant outperforms a general-purpose base model on the tasks Scale-C requires.

To address this question, we evaluated six base-versus-fine-tuned model pairs, ranging from 7B-parameter to 70B-parameter systems, using two independent classification families, natural language inference (NLI) and dense embedding similarity, so that findings do not depend on a single automatic metric. Results are reported across 37 fine-grained cybersecurity topics (grouped under 28 parent categories) and rolled up into eight course-level groups, allowing a detailed view of where fine-tuning helps, where it hurts, and where models generally struggle.

The results reported here correspond to the first evaluation phase (Scale-C Tier 1), which covers multiple-choice knowledge and reasoning. The evaluation of H5P generation, didactic quality, and German localization (Tiers 2 and 3) has not yet been conducted; model recommendations in this report are therefore provisional.

The principal finding is that cybersecurity fine-tuning is not universally beneficial. Two model pairs (Mistral→Lily and Qwen3→Baron) show consistent gains across both classifiers, two others (Llama 3.1→Foundation-Sec and Zephyr→ZySec) show consistent regressions, and the 70B pair (Llama 3.3→Trendyol) yields a classifier-dependent result. Malware and advanced-threat topics remain the most difficult category across all models, indicating a genuine gap in current cybersecurity language modeling rather than a benchmark artifact.

List of Abbreviations

Model Training and Adaptation

Abbreviation	Meaning
CoT	Chain-of-Thought (prompting/reasoning)
DAPT	Domain-Adaptive Pretraining
DPO	Direct Preference Optimization
KL (divergence)	Kullback–Leibler divergence
LLM	Large Language Model
LoRA	Low-Rank Adaptation
NLP	Natural Language Processing
RLHF	Reinforcement Learning from Human Feedback
SFT	Supervised Fine-Tuning

Evaluation and Data

Abbreviation	Meaning
H5P	HTML5 Package (interactive content format used by Scale-C)
MMLU	Massive Multitask Language Understanding (benchmark)
NLE	Non-Linguistic Element (e.g., URLs, file hashes, IP addresses in text)
NLI	Natural Language Inference
RCM	Root Cause Mapping
RQ	Research Question

Cybersecurity Frameworks and Standards

Abbreviation	Meaning
APT	Advanced Persistent Threat
ASVS	Application Security Verification Standard (OWASP)
ATT&CK	Adversarial Tactics, Techniques, and Common Knowledge (MITRE)
CIS	Center for Internet Security
CSF	Cybersecurity Framework (NIST)
CTI	Cyber Threat Intelligence
CVE	Common Vulnerabilities and Exposures
CWE	Common Weakness Enumeration

FedRAMP	Federal Risk and Authorization Management Program (U.S.)
MITRE	MITRE Corporation (developer of ATT&CK and CWE frameworks)
NICE	National Initiative for Cybersecurity Education (U.S. NIST framework)
NIST	National Institute of Standards and Technology (U.S.)
OWASP	Open Worldwide Application Security Project
PCI DSS	Payment Card Industry Data Security Standard
SOC	Security Operations Center

Protocols and Technical Terms

Abbreviation	Meaning
ISO/IEC	International Organization for Standardization / International Electrotechnical Commission
OAuth2	Open Authorization, version 2 (authorization protocol)
OIDC	OpenID Connect (identity layer on top of OAuth2)
SAML	Security Assertion Markup Language
TLS	Transport Layer Security

1. Introduction

1.1 Background and Motivation

Scale-C is a computer security course delivered through interactive H5P units (quizzes, scenarios, and explanations) whose content must be technically correct, classroom-usable, and valid as H5P. A large language model could support this project by drafting activities, writing multiple-choice distractors, explaining concepts, or producing German-language material. The practical question this raises is which model Scale-C should use.

Vendor claims and standalone leaderboard scores are insufficient to answer this question. A model marketed as “cybersecurity fine-tuned” may know domain vocabulary yet select incorrect answers, produce malformed structured output, or handle sensitive prompts poorly; conversely, a general-purpose instruction model may already be adequate. This motivates a controlled benchmarking study rather than reliance on marketing claims.

A growing number of language models advertise a cybersecurity specialization: CyBERTuned, Foundation-Sec, Lily-Cybersecurity, and others, typically on the premise that training on security-specific data improves reasoning about threats, misconfigurations, and defenses. For Scale-C, the relevant question is narrower: when the evaluation setup is held constant, does cybersecurity specialization improve a model on the tasks Scale-C requires, relative to its base model?

“What Scale-C needs” extends beyond multiple-choice recall. The evaluation blueprint for the project distinguishes three tiers of capability: cybersecurity knowledge and reasoning (Tier 1), structured educational output including valid H5P (Tier 2), and optional German localization (Tier 3). This report

addresses Tier 1 in detail; Tiers 2 and 3 are planned but not yet evaluated, and any deployment recommendation should be read as incomplete until those tiers are conducted.

1.2 Research Objective and Guiding Questions

The objective of this study is to determine whether cybersecurity fine-tuning improves language model performance on a broad, topic-diverse cybersecurity benchmark relevant to Scale-C. Rather than ranking cybersecurity models in isolation, each fine-tuned model is compared against its closest base model, matched in size and, where possible, family and serving setup, so that an observed performance gap is more plausibly attributable to the fine-tuning step than to the choice of a different architecture.

The evaluation dataset combines several public benchmark sources rather than a single in-house question bank. Each item is assigned to a Scale-C topic using two classifiers (NLI-based and embedding-based zero-shot labeling), and model performance is reported as multiple-choice accuracy broken down by topic and course group. When the two classifiers disagree on a ranking, this is treated as a stability warning rather than resolved by preference for one method.

This phase of the study addresses the following research questions:

- **RQ1:** Does cybersecurity domain fine-tuning consistently improve benchmark performance, or is the effect pair-dependent?
- **RQ2:** Do NLI-based and embedding-based classifiers agree on model quality, or do they produce different rankings?
- **RQ3:** Does model scale, particularly at 70B parameters, dominate the benchmark, or can well-tuned 7B/8B models compete on selected topics?
- **RQ4:** Does fine-tuning rescue a weak base model, as tested with the Qwen3/Baron pair?

Two further questions are defined for later evaluation phases and are not addressed by the present results: whether leading Tier 1 models can produce valid H5P JSON and usable didactic content (RQ5), and whether top candidates refuse harmful prompts and handle German content acceptably (RQ6).

1.3 Report Structure

Chapter 2 explains why fine-tuning, rather than model branding, is the right lens for this comparison, reviews the training strategies behind the six evaluated pairs, and describes each model individually based on its own published documentation. Chapter 3 describes the methodology, including dataset construction, topic taxonomy, classification procedure, and evaluation design. Chapter 4 reports the results at the aggregate, pairwise, and topic levels. Chapter 5 discusses the findings, states the limitations of the study, and concludes with implications for Scale-C and directions for future work.

2. Theoretical Background

Chapter 1 framed the Scale-C question in practical terms: several models on the market claim a cybersecurity specialization, and the project needs a way to judge whether that claim translates into better performance on the tasks Scale-C actually requires. That specialization is not a marketing label but the product of a specific training step, fine-tuning, applied on top of a general-purpose base model. Before those six pairs can be compared empirically, it is worth being precise about what fine-tuning changes in a model and why the outcome is not guaranteed to be positive; the theory in this chapter is what later motivates treating each pair as a controlled before/after comparison rather than reading cybersecurity models in isolation (Chapter 3).

Fine-tuning can be understood as constrained optimization in a high-dimensional parameter space. Rather than initializing a model from scratch, fine-tuning begins from a pre-trained parameter configuration, denoted $\mathbf{W}_0$, and shifts it toward a target task distribution. This chapter outlines the components of that process and situates the training strategies of the six evaluated model pairs within the literature.

2.1 From Pre-Training to Fine-Tuning

Modern NLP systems are rarely trained from random initialization. A model is first pre-trained on a large general-purpose corpus and subsequently adapted to a downstream task [1]. Fine-tuning reuses the pre-trained weights $\mathbf{W}_0$ as a starting point and updates them on a smaller, task-specific dataset: $\mathbf{W}_{\text{init}} = \mathbf{W}_0$, followed by gradient updates with learning rate η , where η is typically much smaller during fine-tuning than during pre-training [1]. Large update steps risk damaging general capabilities already present in the base model, whereas small steps shift behavior toward the target task without discarding prior knowledge. For Scale-C, this trade-off is concrete: the desired outcome is domain knowledge and stable answer patterns for quizzes and, eventually, H5P generation, without degrading instruction-following or output-format reliability.

2.2 Training Objectives

The model pairs evaluated in this study were adapted using several distinct training objectives, summarized below.

Supervised fine-tuning (SFT)

Given labeled (prompt, target) pairs, training minimizes cross-entropy loss so that the model assigns higher probability to correct tokens. Instruction-tuned models trained on many formatted tasks generalize to new prompts at inference time [3]. Lily and Baron are grouped under this objective according to their published model documentation [7].

Preference optimization (RLHF and related methods)

Some aligned assistants are trained to prefer one response over another: RLHF fits a reward model from human rankings, then fine-tunes the policy while penalizing divergence from a reference model [4]. A risk for Scale-C is that a model optimized this way may sound more helpful or security-aware while becoming less reliable at selecting the correct multiple-choice option; ZySec is grouped as an assistant-style cyber model under this objective [7].

Continued domain pretraining

A second pretraining phase on unlabeled in-domain text often improves downstream performance relative to a general model alone [2]. This typically improves vocabulary and topic coverage but provides no direct supervision on answer format. Foundation-Sec is the primary example of this strategy in the present study [7].

Parameter-efficient adaptation (LoRA)

Full fine-tuning updates every model weight. Low-Rank Adaptation (LoRA) instead freezes $\mathbf{W}_0$ and learns a low-rank update $\Delta\mathbf{W} \approx \mathbf{B} \cdot \mathbf{A}$, reducing memory requirements while matching full fine-tuning on several benchmarks [5]. Most open cybersecurity models in this study use LoRA or a comparable efficient method [5, 7].

KL-style regularization

KL-divergence penalties are common in RLHF pipelines to keep an adapted model close to its base [4]. This step is not universally documented for the cybersecurity models considered here; it is noted as one possible explanation when a fine-tuned model retains domain phrasing yet loses multiple-choice accuracy, as observed for Foundation-Sec and ZySec (Section 4.2).

2.3 Adaptation Strategies and Their Relevance to Scale-C

The six evaluated pairs do not share a single training recipe. Table 2.1 maps each adaptation strategy to its theoretical basis, its expected relevance to Scale-C tasks, and the corresponding model pair.

Table 2.1 – Adaptation strategies and their relevance to Scale-C

Strategy	Basis in literature	Relevance to Scale-C	Pair(s)
Encoder domain adaptation	Cyber-domain BERT pretraining [6]	Baseline only; not suited to generative H5P	RoBERTa → CyBERTuned
Continued domain pretraining	DAPT [2]	Tier 1 factual MCQ	Llama 3.1 → Foundation-Sec
Instruction SFT on cyber data	Instruction tuning [3]	Tier 1 MCQ; Tier 2 H5P (untested)	Mistral → Lily; Qwen3 → Baron
Preference-aligned assistant tuning	RLHF [4]	Learner chat tone; safety (not fully evaluated)	Zephyr → ZySec
Large-scale cyber SFT on a strong base	Instruction tuning [3]	Hard Tier 1 topics; possible Tier 2	Llama 3.3 → Trendyol

CyBERTuned is not a generative Scale-C candidate [6] and is included to isolate how much of the observed performance gap results from architecture (encoder versus decoder) rather than from cybersecurity fine-tuning itself.

2.4 The Six Evaluated Models

Table 2.1 classifies the six pairs by adaptation strategy; this section describes each fine-tuned model individually, as documented in its own model card or accompanying paper. For each pair, three questions are addressed: what the fine-tuning was conceptually intended to achieve, what data and procedure were used to achieve it, and what the model’s own developers reported as the outcome. These developer-reported figures are not comparable to the Phase 1 results in Chapter 4, since they come from different benchmarks and evaluation protocols; they are included here to characterize each model’s design intent rather than to anticipate this study’s findings.

RoBERTa → CyBERTuned

Cybersecurity text is unusually dense with things a language model has never been taught to read properly: URLs, file hashes, IP addresses. Standard masked-language-model pretraining either ignores these non-linguistic elements (NLEs) or treats them as noise to be filtered out. Jang et al. tested several ways of handling them instead and found that a combination of selective masking with a jointly trained NLE-classification objective beat the common approach of stripping NLEs out altogether [6]. CyBERTuned is the model that came out of that comparison: RoBERTa-base, continued for 200 epochs over a cybersecurity corpus using the winning strategy. The original paper reports it outperforming other cybersecurity-domain encoders, including CyBERT and SecureBERT,

on most of six downstream tasks in threat classification and question answering [6]. None of that translates into an ability to generate free text, however. CyBERTuned stays an encoder, so it cannot author Scale-C content on its own; its role here is as an architectural control (Section 3.1), not a deployment candidate.

Llama 3.1 → Foundation-Sec-8B

Cisco's Foundation AI team took a different route with Foundation-Sec-8B: instead of instruction-tuning Llama 3.1 8B, they continued its pretraining on roughly 5.1 billion tokens of security text curated in-house, threat-intelligence reports, vulnerability records, incident-response documentation, security standards, run through a pipeline of web crawling, relevancy filtering, deduplication, and quality control [12]. The idea is to push security vocabulary and concepts into the model's weights before any task-specific tuning happens at all [2]. Cisco evaluated the result on two benchmarks they built for this purpose: CTI-MCQA, 2,500 cybersecurity multiple-choice questions, and CTI-RCM, 1,000 items mapping CVEs to CWE root causes. A later instruction-tuned version reportedly beat both Llama 3.1-8B-Instruct and GPT-4o-mini on CTIBench-RCM. It is worth noting what this training recipe is not: it is domain pretraining, not instruction tuning, and that distinction lines up with what Chapter 4 finds for this pair. A model can pick up domain vocabulary without getting any better at picking the labeled correct answer on a multiple-choice question.

Mistral → Lily-Cybersecurity-7B-v0.2

Lily goes the opposite direction from Foundation-Sec. Rather than continued pretraining on raw text, it is an instruction fine-tune of Mistral-7B-Instruct-v0.2 built on 22,000 hand-crafted cybersecurity and hacking question-answer pairs, which were then run back through a large language model to add conversational tone and a consistent persona. Its developers trained it for five epochs on a single A100, about 24 hours of compute according to the public model card [13]. There is no independently published quantitative benchmark for Lily; its release notes describe intended use cases (threat analysis, incident-response planning, security-awareness training) rather than reporting a score. That informality makes what shows up in Chapter 4 somewhat notable: Lily is the clearest positive fine-tuning result of the six pairs in this study's own evaluation (Section 4.3). A relatively small, hand-curated Q&A dataset appears to have been enough to move the needle, even without a formal benchmark from the model's own authors to back that up in advance.

Zephyr → ZySec-7B

ZySec-AI describes ZySec-7B less as a knowledge base and more as an always-available security teammate. Training covers conversational usefulness across more than 30 domains, attack-surface analysis, cloud security, compliance frameworks such as CIS Controls, FedRAMP, PCI DSS, and ISO/IEC 27001 [14]. Built on Zephyr-7B-Beta, the model is trained with Direct Preference Optimization (DPO), a preference-based alignment method in the same family as RLHF: it fits the model toward preferred responses over dispreferred ones, rather than toward a single labeled correct answer [4]. As with Lily, no independently published benchmark accompanies the release; the documentation emphasizes domain breadth and conversational role instead of a score. That training objective, optimizing for helpfulness and preferred phrasing rather than for picking one correct label, may explain what Chapter 4 finds: ZySec shows the largest consistent regression of any pair in this study's MCQ evaluation (Section 4.3).

Qwen3 → BaronLLM v2

BaronLLM v2 is built for a narrower purpose than the other fine-tunes in this study: offensive-security research, red-team scenario generation, exploit reasoning, attack-chain construction, developed by the Trendyol Group Security Team under stated safety constraints meant to keep disallowed content out. The fine-tuning itself is supervised instruction tuning with chain-of-thought supervision on top of

Qwen3-14B, using 53,202 curated examples across more than 200 cybersecurity subdomains. Copyrighted exploit code and raw shellcode or binary payloads were explicitly excluded from the training set [15]. Training took roughly 65 hours on four NVIDIA H100 GPUs. The developers report about a 1.5× improvement over the Qwen3-14B base on the CyberSec-Eval benchmark, along with an informal placement on an external leaderboard (13th overall, 4th among comparably sized public models). Those figures cannot be compared directly against Chapter 4’s results, since CyberSec-Eval scores different content in a different way. But the direction is consistent: BaronLLM shows a sizable gain over its own base in this study too (Section 4.3), even though that base, Qwen3-14B, scores unusually low here in absolute terms.

Llama 3.3 → Trendyol Cybersecurity LLM v2 70B

The largest model in this study comes from the same Trendyol Group Security Team, but with a different brief: defense-oriented consulting rather than offensive research. Its intended uses are enterprise defense consulting, architectural guidance, control-framework mapping, and threat-intelligence support, and it is built to refuse unauthorized penetration-testing requests or live exploit generation outright. The training recipe is supervised instruction tuning combined with system-prompt-level guardrails on top of Llama 3.3 70B Instruct. The dataset behind it grew roughly fourfold between versions, from 21,258 rows in v1.1 to 83,920 in v2.0, spanning OWASP, MITRE ATT&CK, NIST CSF, CIS Controls, the Australian Signals Directorate’s Essential Eight, plus modern identity protocols (OAuth2, OIDC, SAML), TLS, cloud/DevSecOps practice, and cryptography [15]. The developers report a near-top placement on the CS-Eval benchmark in both English and English–Chinese tracks. This is not directly comparable to the results in Chapter 4. But the emphasis on broad framework coverage under an SFT-plus-guardrails regime lines up reasonably well with the mixed result this study finds for the pair — with one caveat: the base-model NLI run is incomplete (Section 4.3).

3. Methodology

3.1 Evaluation Design

Ranking cybersecurity models in a single table would not distinguish whether a high score originates from fine-tuning, from a larger base model, or from a different tokenizer and chat template. Pairing each adapted model with its closest base model controls for these confounds [1, 7]. Table 3.1 lists the six evaluated pairs; all generative pairs follow the same Phase 1 protocol: identical items, identical decoding settings, identical MCQ-extraction logic.

Table 3.1 – Base and fine-tuned model pairs

Base model	Fine-tuned / domain model	Role in study
RoBERTa base	CyBERTuned	Encoder baseline [6]
Llama 3.1 8B	Foundation-Sec-8B	Domain-pretrained LLM [2, 7]
Mistral-7B-Instruct-v0.2	Lily-Cybersecurity-7B-v0.2	Instruction-tuned cyber model [3, 7]
Zephyr-7B-Beta	ZySec-7B	Aligned cyber assistant [4, 7]
Qwen3 14B	BaronLLM v2	Instruction-tuned cyber model on mid-size base [3, 7]
Llama 3.3 70B Instruct	Trendyol Cybersecurity LLM v2 70B	Large instruct model + cyber SFT [3, 7]

The encoder pair (RoBERTa → CyBERTuned) is a special case, scored by similarity ranking rather than text generation [6], and serves as a scientific control rather than a deployment candidate. The 7B–8B pairs represent a realistic deployment scale for a university project in terms of GPU cost and hosting; the Qwen3/Baron pair tests whether fine-tuning can compensate for a weak base model; and the 70B pair tests whether cybersecurity SFT adds value when the base model is already strong. Beyond accuracy, Scale-C will eventually require valid H5P JSON output (Tier 2), acceptable refusal behavior on exploit-style prompts, German or localized content, and compatibility with available hardware and licensing constraints; Phase 1 does not evaluate these dimensions, so Table 3.1 should be read as a shortlist for further testing rather than a final selection.

3.2 Dataset Construction

The evaluation dataset is not Scale-C course content but a pooled set of public benchmark items selected for topical overlap with the Scale-C curriculum, used to compare candidate models prior to platform integration. An earlier pipeline snapshot processed approximately 1,777 records, of which 961 were scorable multiple-choice items; final counts for a given run should be taken from the current manifest in the project repository [11].

The eight sources in Table 3.2 were not the only cybersecurity benchmarks available; they were selected against three criteria. First, coverage breadth: sources were chosen so that, jointly, they touch every course group in the Scale-C taxonomy (Table 3.3) rather than concentrating on a single sub-area such as malware analysis or network defense. Second, scoring compatibility: Phase 1 requires an unambiguous gold answer, so priority was given to benchmarks already structured as multiple-choice or otherwise machine-scorable items, deferring free-text or agentic benchmarks to later phases. Third, provenance and accessibility: all eight sources are public, versioned, and retrievable through Hugging Face Datasets, which keeps the pipeline reproducible and avoids relying on proprietary or paywalled item banks. Benchmarks that failed on any of these three criteria, for example sources limited to a narrow sub-topic, or distributed only as unscored raw text, were excluded from this phase and remain candidates for later, more targeted evaluation.

Table 3.2 lists the benchmark sources used to construct the dataset, together with the cybersecurity and educational frameworks (MITRE ATT&CK, NIST CSF 2.0, NICE, CWE, OWASP ASVS, Bloom's taxonomy) that describe their coverage. These frameworks describe how coverage is conceptualized; they are not the automatic topic labels used by the classification pipeline.

Table 3.2 – Benchmark sources and associated frameworks

Benchmark source	Main contribution	Associated framework	Relevance
MMLU (computer security)	General computer-security knowledge and MCQ reasoning	NICE; revised Bloom's taxonomy	Measures knowledge and cognitive level rather than adversary technique
CyberBench	Cybersecurity knowledge, threat analysis, reasoning	MITRE ATT&CK; NIST CSF 2.0; NICE	Covers attacker behavior, defensive outcomes, practitioner skills
CyberMetric	Expert-validated cybersecurity knowledge	NIST CSF 2.0; NICE	Standards-oriented cybersecurity knowledge and practice
CyberSOCEval	SOC tasks, threat intelligence, malware analysis	MITRE ATT&CK; NIST CSF 2.0; NICE	SOC reasoning maps to adversary tactics, detection, response, work-role skills

SEC-bench	CVE, open-source, application-security evaluation	MITRE CWE; OWASP ASVS	Software weaknesses, vulnerabilities, application-security verification
Global-MMLU	Multilingual and localized knowledge evaluation	NICE; revised Bloom's taxonomy	Extends knowledge evaluation across languages, including German
JSONSchemaBench	Structured output generation	JSON Schema	Evaluates schema-conformant structured output
superGLEBer	German-language and multilingual NLP capability	Scale-C multilingual/localization tier	Supports German and multilingual behavior rather than a cybersecurity framework directly

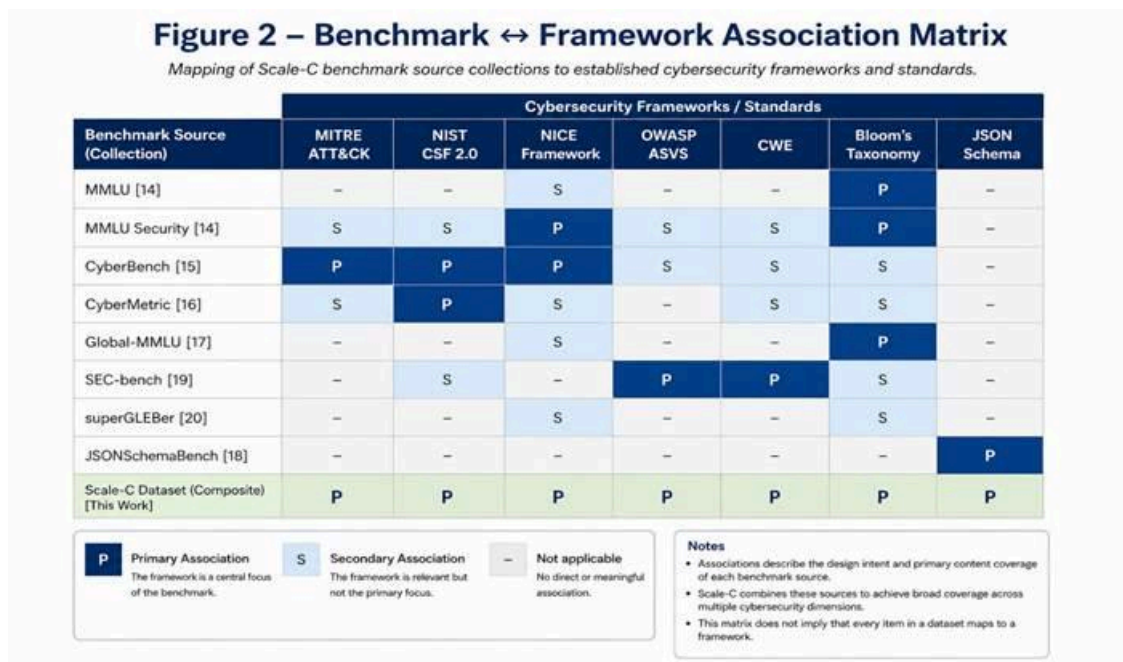


Figure 1. Benchmark–Framework Association Matrix.

3.3 Topic Taxonomy

Items are mapped to Scale-C topics defined in the project's taxonomy schema, which contains 37 leaf labels (e.g., email, SSL certificates, advanced malware types) grouped under 28 parent codes (SCLC001–SCLC028). For course-level reporting, these leaves are further bundled into eight curriculum groups, listed in Table 3.3. Proposed metadata fields in the schema (risk, difficulty, and cognitive skill) are not yet validated and are not used in the Phase 1 results.

Table 3.3: Eight course groups (curriculum-level rollout)

Course group	Example leaf topics
Identity and access	Credential management, authentication/authorization, unique logins, encrypted vaults
Human factors and social engineering	Social engineering, email, psychological manipulation, behavioral security

Malware and advanced threats	Viruses/Trojans, adware, advanced malware types, advanced cyber threats
Network and secure communications	Firewalls, SSL certificates, secure communication, infrastructure protection
Safe computing and browsing	General hygiene, safe browsing, USB safety, header checks, typos, domains
Governance and organizational risk	Regulatory/legal, internal risks, professional perspective
Application and distributed work	App security, working from home, remote/traveling security
Foundations, defense, and resilience	Foundational concepts, general theory, defensive measures, backups & recovery, other

3.4 Topic Classification

Each item is labeled independently by two zero-shot methods. Under NLI labeling, item text is scored against each candidate topic description as an entailment problem [9], using a multilingual NLI classifier (MoritzLaurer/mDeBERTa-v3-base-xnli-multilingual-nli-2mil7). Under embedding labeling, the item and each topic description are embedded, and the closest label by cosine similarity is assigned [10]. These classifiers assign topics to questions rather than scores to model answers; running both methods produces two topic-labeled views of the same item pool, and a ranking that changes between labelers is treated as a stability warning (RQ2). After labeling, the dataset builder deduplicates items, retains valid predictions, and caps the number of items any single source may contribute per topic, so that no single large dataset dominates the pool [11].

3.5 Evaluation Metric and Pipeline

The evaluation step operates on the classified and balanced dataset. Each model receives the same benchmark item under the same scoring configuration; for multiple-choice questions, model output is compared with the gold answer, and the primary metric is accuracy, defined as the number of correct answers divided by the number of scorable questions. This is currently the most reliable automatic metric because MCQ items have an unambiguous gold answer. Non-MCQ records, open explanations, structured H5P generation, German localization, or extended reasoning, require a different evaluation method and are left to a later phase employing a rubric, human evaluation, or a validated judge model.

The primary comparison is pairwise: each fine-tuned cybersecurity model is compared with its closest base model, and a fine-tuning delta is computed as the accuracy of the fine-tuned model minus the accuracy of the base model. A positive delta indicates that the fine-tuned model outperformed its base; a negative delta indicates the reverse. Figure 2 summarizes the end-to-end pipeline, from raw benchmark sources to taxonomy-aware model evaluation and category-level analysis.

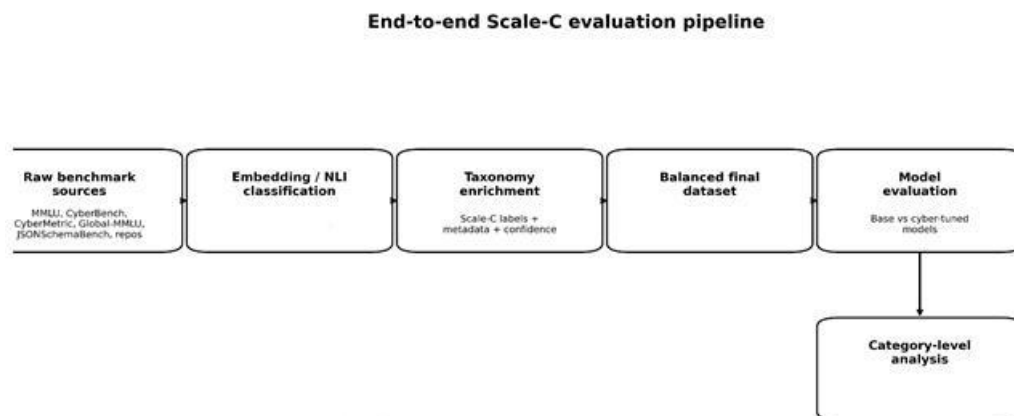


Figure 2. Pipeline from raw benchmark sources to taxonomy-aware model evaluation and category analysis.

4. Results

The results below move from dataset to leaderboard to detail. Section 4.1 describes the benchmark dataset itself, since every accuracy figure that follows depends on how it was built. Section 4.2 gives the overall Tier 1 leaderboard, Section 4.3 breaks it down pair by pair, and Section 4.4 goes further still, down to individual topics. All numbers derive from the repository run stored in `data/eval_1/` (evaluation corpus `data/final_1/`, classifier-specific JSONL inputs); any figure from an earlier presentation that conflicts with this run should be treated as superseded.

4.1 Benchmark Dataset Overview

Section 3 describes how the evaluation set is built. What follows are the concrete Phase 1 counts that underlie every accuracy in this chapter, replacing the provisional snapshot noted in Section 3.2.

After classification and gold-answer filtering (Sections 3.4–3.5), the NLI-labeled pool yields 897 classified items, of which 573 are scorable; the embedding-labeled pool yields 1,579 classified items, of which 847 are scorable. These two denominators are the ones used in Table 4.2 for all fully completed runs. They are not identical item sets: independent topic assignment produces different mixes and different per-topic support, which is why results are reported as two views rather than one merged ranking.

Of the 573 NLI-scorable items, 322 are declared multiple-choice in the source metadata and 251 are typed as open explanation but still carry A–D options with a gold letter; both are scored by exact match. The embedding view follows the same pattern ($549 + 298 = 847$). Sources retained in the broader classified pool but contributing no Tier 1 gold answers—chiefly JSONSchemaBench, superGLEBer, SEC-bench, and parts of CyberBench—are held for later structured-output and localization work rather than discarded.

Table 4.1a – Phase 1 evaluation pools (scorable Tier 1)

Dimension	NLI-classified corpus	Embedding-classified corpus
Classified items available for evaluation	897	1,579

Scorable items (Table 4.2 denominator)	573	847
Leaf topics covered (of 37)	32	37
Parent codes covered (of 28)	27	28
Course groups covered (of 8)	8	8
English / German (scorable)	487 / 86	644 / 203

Table 4.1b – Scorable items by benchmark family

Dimension	NLI-classified corpus	Embedding-classified corpus
CyberMetric	251	298
CyberSOCEval (incl. threat-intelligence and malware splits)	146	126
Global-MMLU	86	316
MMLU (computer security / college computer science)	90	107
Total	573	847

Table 4.1c – Scorable items by course group

Dimension	NLI-classified corpus	Embedding-classified corpus
Fundamentals and governance	124	132
Human and social threats	84	177
Defensive controls and hygiene	77	112
Network and secure communications	77	76
Malware and advanced threats	72	112
Identity and access	61	82
Applications and workplace	48	132
Other	30	24
Total	573	847

Coverage across the eight course groups is complete, but leaf-level support is uneven. Under NLI labeling, five topics have no scorable items (USB safety, friend requests, unique

logins, default usernames, typos). The embedding view covers all 37 leaves, yet unique logins (n = 2) and USB safety (n = 5) remain too thin for reliable topic claims; further sparse cells are handled as in Section 4.4.

One modeling caveat affects denominators directly. Llama 3.3 70B Instruct was scored on a later, topic-capped build rather than the shared 573 / 847 pools, and its NLI run stopped at 327 scorable items. BaronLLM v2 is also short by a few items (569 / 573 NLI; 837 / 847 embedding) after transient API failures. These exceptions are footnoted under Table 4.2 and revisited in Section 5.4; until the 70B pair is re-evaluated on the same frozen item set, its deltas should not be over-interpreted.

4.2 Overall Tier 1 Leaderboard

Twelve systems were evaluated: six base/fine-tuned pairs from Table 3.1. The NLI-classified corpus contains 573 scorable items per generative model (encoder pair included); the embedding-classified corpus contains 847 scorable items per generative model, with the exception of Llama 3.3 70B (see notes below Table 4.2). Because the NLI and embedding views assign topic labels to a partially different item pool, they are reported as two complementary views rather than combined into a single ranking.

Table 4.2 – Phase 1 overall MCQ accuracy

Model	Role	NLI acc. (n/N)	Embedding acc. (n/N)
RoBERTa base	Encoder baseline	21.1% (121/573)	26.7% (226/847)
CyBERTuned	Encoder fine-tuned	21.6% (124/573)	27.0% (229/847)
Llama 3.1 8B	Base	57.9% (332/573)	60.9% (516/847)
Foundation-Sec-8B	Fine-tuned	52.2% (299/573)	52.2% (442/847)
Mistral-7B-Instruct-v0.2	Base	44.5% (255/573)	46.9% (397/847)
Lily-Cybersecurity-7B	Fine-tuned	51.5% (295/573)	53.7% (455/847)
Zephyr-7B-Beta	Base	37.9% (217/573)	43.2% (366/847)
ZySec-7B	Fine-tuned	27.1% (155/573)	36.0% (305/847)
Qwen3 14B	Base	11.9% (68/573)	13.9% (118/847)
BaronLLM v2	Fine-tuned	26.2% (149/569)	30.8% (258/837)
Llama 3.3 70B Instruct	Base	55.7% (182/327)†	75.7% (1024/1353)‡
Trendyol Cybersecurity LLM v2 70B	Fine-tuned	65.1% (373/573)	68.5% (580/847)

† Partial NLI run: Llama 3.3 70B base was evaluated on 327 of 573 NLI items at the time these results were generated; NLI rankings involving this model should be treated as provisional until the run is completed. ‡ Embedding denominator difference: the Llama 3.3 70B embedding evaluation includes 1,353 scorable items, reflecting additional task types beyond strict MCQ within the same corpus, whereas other models use 847 items; embedding accuracy for this model should not be compared directly against 847-item models without harmonizing the item set.

Within the NLI view (573 items), Trendyol 70B leads at 65.1%, followed among fully evaluated generative models by Llama 3.1 8B (57.9%) and Lily (51.5%); Qwen3 14B is weakest at 11.9%. Within the embedding view (847 items, comparable n), Trendyol 70B again leads at 68.5%, followed by Lily (53.7%) and Foundation-Sec (52.2%), with Qwen3 14B again lowest at 13.9%. There is no single global leader across classifiers: NLI and embedding views can reorder models because topic assignment and item counts differ [9, 10], a discrepancy discussed further in Chapter 5 for the 70B pair. A high absolute score in Table 4.2 supports continued evaluation, Tier 2/3 testing, safety checks, hosting cost, but does not by itself constitute a deployment recommendation.

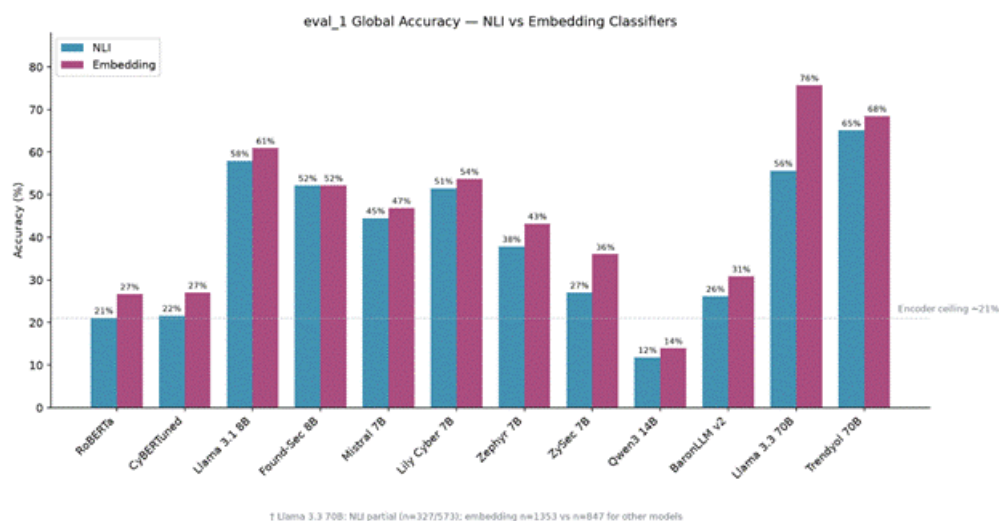


Figure 3. Global accuracy under NLI versus embedding topic labeling for all twelve systems. Dashed line marks the encoder ceiling ($\approx 21\%$ NLI); Llama 3.3 70B NLI is partial (327/573) and embedding n varies (see Table 4.2).

Encoder ceiling versus generative models

The encoder pair (RoBERTa $\rightarrow$ CyBERTuned) scores 21–27% regardless of classifier [6], with CyBERTuned improving only marginally over RoBERTa (+0.5 pp NLI, +0.3 pp embedding), an expected outcome, since both models answer by ranking choices via embeddings rather than generating text. The transition to a 7B generative instruction model (Llama 3.1 8B) adds approximately 37 pp under NLI and 34 pp under embedding, a gap that dominates any gain attributable to cybersecurity encoder fine-tuning. For Scale-C's intended use cases, drafting quiz text, producing structured H5P, explaining concepts to learners, this indicates that generative instruction-tuned LLMs are the relevant candidate class, with encoder pairs serving only as a scientific control.

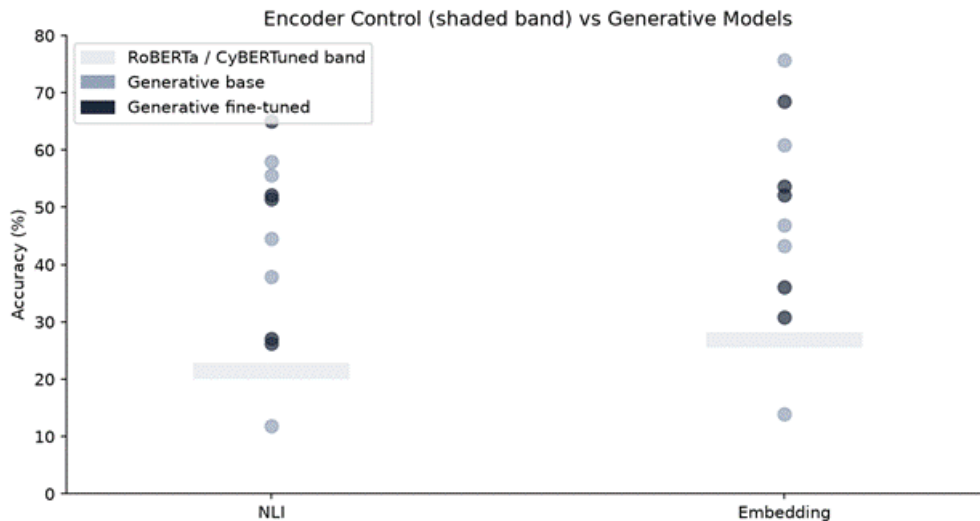


Figure 4. Encoder control (shaded band) versus generative models at 7B–8B and 70B scale.

Interpretation of the aggregate results

A cybersecurity label on a model card does not guarantee better accuracy: Foundation-Sec and ZySec both score below their own base models here, so branding alone predicts little about Tier 1 performance [1, 7]. Scale clearly helps, but it is not the whole story either: the 70B models lead both views (with the denominator caveats noted above), yet Llama 3.1 8B and Lily stay competitive in absolute terms at a fraction of the hosting cost. Even a weak base is not necessarily a lost cause. Qwen3 14B is the weakest model in the study by a wide margin, and BaronLLM v2 still roughly doubles that score (Section 4.2). What does not improve, for any model, is malware and advanced-threat performance (Section 4.3): content pipelines built around those topics should assume human review is needed, not optional.

These aggregate figures do not address valid H5P JSON output or didactic quality (Tier 2), German or multilingual behavior (Tier 3), safety refusals on exploit-style prompts, or pairwise fine-tuning effects and topic-level strengths, which are addressed in the remainder of this chapter. Table 4.2 should accordingly be read as a Phase 1 screening result rather than a final model choice.

4.3 Pairwise Fine-Tuning Effects

The aggregate leaderboard in Table 4.2 obscures an important pattern: several cybersecurity-branded models score below their own base model once pairs are compared directly. Table 4.3 groups the six pairs into three outcome categories.

Table 4.3 – Fine-tuning outcome categories

Category	Pairs	Δ NLI / Δ Embedding	Interpretation
Clear improvement	Mistral → Lily; Qwen3 → Baron	+7.0 / +6.8 pp; +14.3 / +16.9 pp	Fine-tuning clearly improved performance; Lily is a credible 7B shortlist candidate, while Baron rescues a weak base but remains mid-pack in absolute terms.

Clear regression	Llama 3.1 → Foundation-Sec; Zephyr → ZySec	-5.8 / -8.7 pp; -10.8 / -7.2 pp	The base models outperform their fine-tuned counterparts on Tier 1 MCQ; these fine-tuned variants would not be selected over their bases without new evidence from later tiers.
Mixed or marginal	RoBERTa → CyBERTuned; Llama 3.3 → Trendyol	+0.5 / +0.4 pp; +9.4 / -7.2 pp	The encoder gain is negligible; the 70B pair reverses sign between classifiers and should be treated as inconclusive until the base NLI run is completed on a shared item set.

Table 4.4 – Pairwise fine-tuning deltas (data/eval_1)

Pair	Training style	Base (NLI / Emb.)	Fine-tuned (NLI / Emb.)	Δ NLI	Δ Emb.
RoBERTa → CyBERTuned	Encoder domain SFT [6]	21.1% / 26.7%	21.6% / 27.0%	+0.5 pp	+0.4 pp
Llama 3.1 → Foundation-Sec	Domain pretraining [2]	57.9% / 60.9%	52.2% / 52.2%	-5.8 pp	-8.7 pp
Mistral → Lily	Instruct SFT [3]	44.5% / 46.9%	51.5% / 53.7%	+7.0 pp	+6.8 pp
Zephyr → ZySec	Assistant tuning [4]	37.9% / 43.2%	27.1% / 36.0%	-10.8 pp	-7.2 pp
Qwen3 → Baron	Instruct SFT [3]	11.9% / 13.9%	26.2% / 30.8%	+14.3 pp	+16.9 pp
Llama 3.3 → Trendyol	Large-scale SFT [3]	55.7%† / 75.7%‡	65.1% / 68.5%	+9.4 pp	-7.2 pp

† Llama 3.3 base NLI is a partial run (182/327 of 573 items). ‡ Embedding denominators differ (base 1,353 items vs. fine-tuned 847 items; see Section 4.1). The central implication for model selection is that a “cybersecurity fine-tuned” designation on a model card is not evidence of improved benchmark performance [7]: two pairs improved on both views, two regressed on both views, and the remainder require closer interpretation.

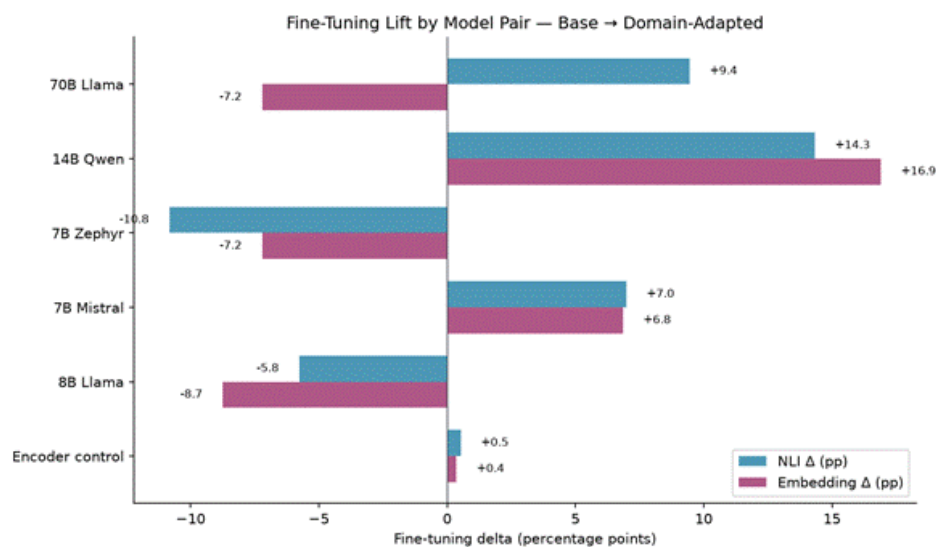


Figure 5. Fine-tuning delta by pair, under NLI and embedding topic views. The Llama 3.3 → Trendyol row is the only pair with opposite signs.

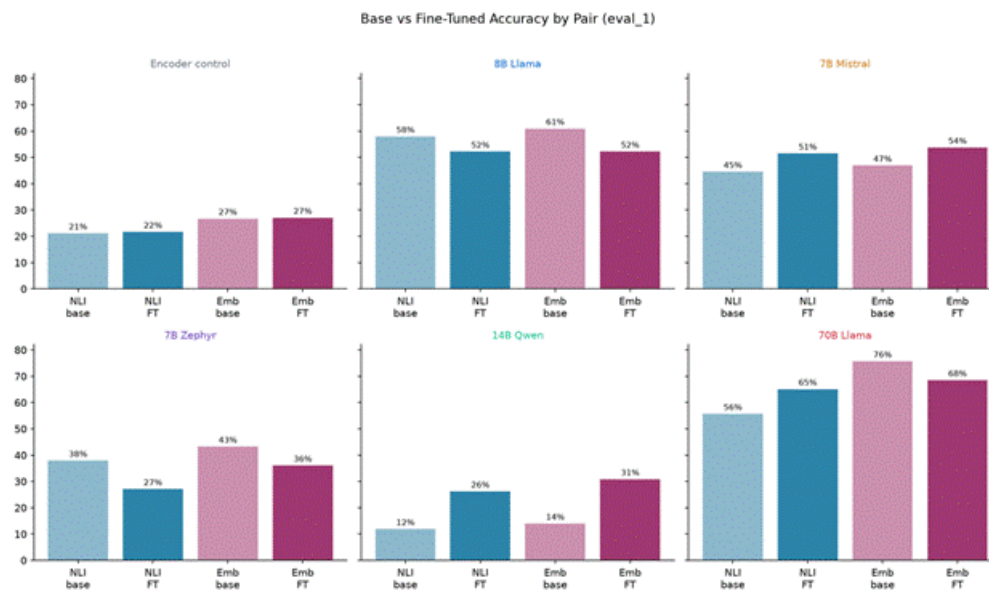


Figure 6. Base versus fine-tuned accuracy within each pair, under both classifier views.

Pair-level observations

The RoBERTa → CyBERTuned pair adds approximately half a percentage point on each view [6], a difference that is statistically negligible and does not alter the encoder-versus-generative gap identified in Section 4.1; this pair functions only as a sanity check, not as a Scale-C candidate.

Foundation-Sec, trained via continued domain pretraining on security text [2, 7], was expected to improve vocabulary and topic coverage, yet Tier 1 MCQ accuracy drops by 5.8 pp (NLI) and 8.7 pp (embedding) relative to Llama 3.1 8B. One explanation, echoing the KL-forgetting discussion in Chapter 2 [4], is that the model absorbed domain phrasing while disrupting answer patterns the base model had already learned for multiple-choice prompts; on Phase 1 evidence, Llama 3.1 8B is the more reliable choice in this family pending evidence from later tiers.

Lily is the clearest success story among the 7B-class models, with both classifiers showing a gain of approximately 7 pp; the fine-tuned model reaches 51.5% (NLI) and 53.7% (embedding), not the global leader, but a substantial improvement over a mid-40s base. For a project that may require modest-GPU hosting, Mistral → Lily provides the strongest evidence that fine-tuning was worthwhile among the smaller generative pairs, and it remains a priority candidate for Tier 2 H5P testing.

ZySec shows the largest consistent regression (−10.8 pp NLI, −7.2 pp embedding); its base, Zephyr, is already comparatively weak on this benchmark, and the fine-tuned model performs worse still (27.1% NLI). Assistant-style cybersecurity tuning [4] may optimize for conversational tone at the expense of MCQ reliability, and ZySec should not be assumed to improve on Zephyr for quiz-related tasks.

BaronLLM v2 shows a large apparent gain (+14.3 / +16.9 pp), but this must be read against an unusually low base score for Qwen3 14B (11.9% NLI, 13.9% embedding), which likely reflects a mismatch between Qwen3’s default chat template and the MCQ-extraction pipeline rather than a general weakness of Qwen3 [7]. Baron roughly doubles the base score, reaching 26–31% overall, but remains below the top absolute performers in Table 4.2; fine-tuning here appears to correct a poorly served baseline more than it establishes a new leader, and Baron is of interest mainly as a mid-size candidate contingent on confirming that the Qwen3 base was mis-served.

The Llama 3.3 → Trendyol pair is the least conclusive result. Under NLI, Trendyol leads the entire study at 65.1% and exceeds its base by 9.4 pp, but the base NLI run is incomplete (327 items); under

embedding, Trendyol remains strong at 68.5% in absolute terms yet loses 7.2 pp to the base because Llama 3.3 was scored on a larger, mixed-task corpus (1,353 items). No recommendation of Trendyol over Llama 3.3 can be drawn from these deltas alone; a completed, same-item re-evaluation of both models is required before any 70B deployment decision.

If the field had to be narrowed today, using only Tier 1 MCQ and the pairwise evidence collected so far, Lily and the Llama 3.1 8B base would warrant continued investigation, along with the Llama 3.3/Trendyol line once a clean re-run is available. Foundation-Sec, ZySec, and encoder-only paths should be deprioritized for MCQ purposes, and Baron and Trendyol should be treated with caution pending further evidence. None of these conclusions extend to H5P JSON validity, German content, or safety, all of which remain untested. This answers RQ1: the effect of cybersecurity fine-tuning is pair-dependent. Two pairs gain on both classifiers, two lose on both, one is marginal, and one reverses sign, so there is no general rule that cybersecurity fine-tuning improves Scale-C-style MCQ performance [1–4].

4.4 Course-Level and Topic-Level Patterns

Global accuracy figures conceal patterns relevant to Scale-C: a model can perform adequately overall while failing on malware questions, phishing scenarios, or regulatory content, topics taught in distinct H5P modules.

Under the embedding view, the strongest-performing models cluster on course groups that reward factual, well-represented knowledge, as shown in Table 4.5 for identity/access and network/secure-communications groups, which cover credentials, authentication, firewalls, TLS, and secure communication: topics well represented in general pretraining corpora and standard security curricula [1].

Table 4.5 – Strongest course groups (embedding view, selected models)

Course group	Llama 3.3 70B	Trendyol 70B	Llama 3.1 8B	Lily 7B
Identity & access	89.4% (n=151)	68.4% (n=82)	72.0% (n=82)	63.4% (n=82)
Network & secure comms	89.5% (n=124)	81.6% (n=76)	69.7% (n=76)	63.2% (n=76)

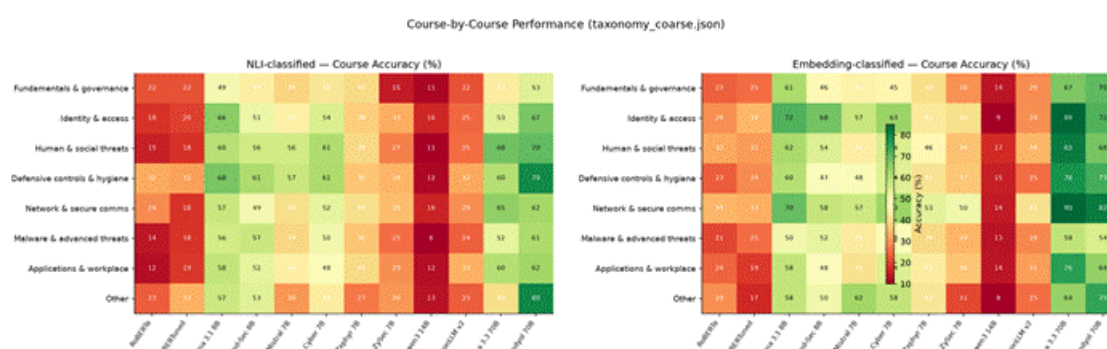


Figure 7. Course-level accuracy heatmap across eight course groups and twelve models, split into NLI-classified (left) and embedding-classified (right) corpora. Malware/advanced-threats and fundamentals rows remain darker across models; identity and network rows brighten for larger models.

At the fine-topic level, three topics recur as persistent weaknesses. Table 4.6 reports NLI/embedding accuracy for four representative models on the same fine labels, with support counts in parentheses.

Table 4.6 – Difficult fine topics (NLI% / embedding%)

Topic	Qwen3 14B	BaronLLM v2	Llama 3.1 8B	Llama 3.3 70B
General theory	8% / 6% (40/31)	21% / 26% (39/31)	48% / 58% (40/31)	20% / 58% (20/65)†
Advanced cyber threats	7% / 16% (15/32)	20% / 28% (15/32)	40% / 62% (15/32)	55% / 50% (11/60)†
Advanced malware types	7% / 9% (29/33)	28% / 16% (29/32)	69% / 39% (29/33)	59% / 58% (17/67)†

† Llama 3.3 NLI counts reflect the partial run (Section 4.1); embedding denominators also differ for this model. Theory and advanced-threat topics reward scale and broad pretraining, with Llama 3.1 8B often outperforming Qwen3 and Baron by a wide margin. Advanced malware results are mixed: under NLI, Lily and Llama 3.1 reach approximately 69% on their slices (unusually high for this topic), while embedding scores stay lower for most 7B models. Format and classifier view both shape the outcome, so no single number should guide curriculum planning. Qwen3’s weakness is broad rather than concentrated in one topic, which points to a serving or template issue that fine-tuning only partially corrects.

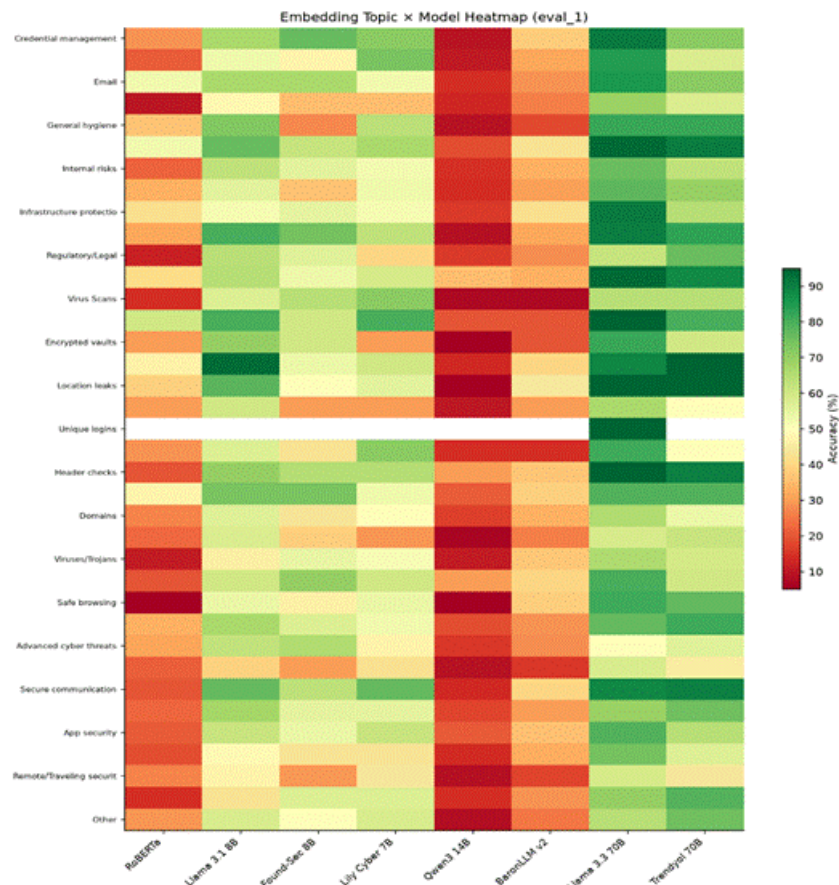


Figure 8. Fine-topic accuracy by model, embedding view. Malware and threat rows remain consistently dark; identity, network, and SSL-related rows brighten for larger or better-matched models; the Qwen3 column is uniformly weak.

Implications for curriculum-model matching

- Identity, network, and certificate-style MCQs are the most reliable candidates for LLM-assisted drafting in Phase 1; malware, advanced threats, and broad theory require more human oversight.

- No model performs uniformly well across topics: Lily helps on malware but can underperform on email content, and Trendyol gains on some NLI slices while losing on social engineering under embedding.
- Topic-level findings should always be reported together with the classifier view (NLI or embedding [9, 10]), since a gain under one view may not appear under the other.
- Topics with fewer than 10 support items (e.g., general hygiene, USB safety) can show swings of ± 40 pp that reflect noise as much as signal; topics with $n \geq 15$ provide more reliable evidence for curriculum decisions.
- These results do not establish whether strong MCQ performance on a topic translates into valid H5P JSON (Tier 2) or correct German-language items (Tier 3).

5. Discussion

The point of this study was never to crown a cybersecurity leaderboard winner. It was to decide, for Scale-C specifically, whether fine-tuning is worth trusting. Phase 1 only covers Tier 1: multiple-choice knowledge and reasoning, so H5P generation, German localization, and safety checks are still open questions. Read the conclusions below with that in mind.

5.1 Answers to the Research Questions

RQ1 — does cybersecurity fine-tuning consistently help? No; the effect depends on the pair. Lily (+7 pp) and Baron (+14–17 pp) improve on both classifier views; Foundation-Sec (−6 to −9 pp) and ZySec (−7 to −11 pp) regress; Trendyol is inconclusive (+9.4 pp NLI, −7.2 pp embedding). A cybersecurity label on a model card is not a reliable predictor of Tier 1 performance [1, 7].

RQ2 — do rankings remain stable across topic labelers? Largely, but not where it matters most: overall rank order correlates strongly across classifiers ($\rho \approx 0.97$), yet the overall leader reverses (Trendyol under NLI, Llama 3.3 under embedding) and the 70B fine-tuning verdict changes sign. A single automatic topic-labeling pipeline should not be relied upon for model selection [9, 10].

RQ3 — do 70B models dominate? On many slices, yes, but not universally. Llama 3.3 and Trendyol lead on identity, network, and absolute MCQ score, while well-tuned 7B models, particularly Lily and the Llama 3.1 8B base, remain within range for a resource-limited deployment. Scale improves performance on hard theory/threat topics but does not resolve every course group; even Llama 3.3 70B, the strongest model in the study, scores only 58–59% on advanced malware and as low as 20% (NLI) on general theory (Table 4.6).

RQ4 — can fine-tuning rescue a weak base? Partially: Baron roughly doubles Qwen3’s score but remains mid-pack overall. Fine-tuning looks to have corrected a poorly matched base more than it produced a new leader. For Scale-C, this suggests addressing serving and prompting configuration before evaluating whether cybersecurity SFT is worth its cost.

5.2 Principal Findings

An aggregate leaderboard mixes models that improved under fine-tuning with models that regressed. That is the main reason a pairwise comparison is the right unit of analysis here: only a direct base-versus-fine-tuned reading isolates what adaptation itself actually did (Section 4.3). It is also why generative models, not encoders, are the relevant candidate class for Scale-C, encoder fine-tuning added roughly 0.5 pp in this study, while simply moving to a 7B generative model added roughly 37 pp (Section 4.2). H5P authoring needs generation, not choice ranking, so the encoder path was never really in contention.

Aggregate numbers also hide where a model actually helps. Lily gains on malware but can underperform on email; Trendyol gains on some NLI slices but loses on social engineering under embedding (Section 4.4). Put differently: no model here is simply “good at cybersecurity”; that claim only means something once topic and classifier are specified. Malware and advanced-threat content is the clearest example of a gap that does not go away with model choice. Every system evaluated underperforms there relative to identity and network content, which alone is reason to keep a human in the loop for those modules regardless of which model Scale-C picks.

A smaller point, but one that shaped the whole design: running two classifiers instead of one was not a formality. If either view alone were sufficient, NLI and embedding would agree with each other, and for the 70B pair, they do not (Section 4.3).

5.3 Implications for Model Selection

These results constitute a shortlist and an avoid-list for Tier 2/3 testing, not a final deployment decision. Table 5.1 summarizes the recommended status of each candidate on the basis of Phase 1 evidence.

Table 5.1 – Recommended status for Tier 2 testing

Status	Candidate(s)	Rationale
Continue evaluating	Lily-Cybersecurity-7B	Clearest 7B fine-tuning gain; reasonable absolute MCQ accuracy; realistic GPU budget for a university deployment.
Continue evaluating	Llama 3.1 8B (base)	Outperforms Foundation-Sec on Tier 1; strong on theory/threat slices relative to Qwen3/Baron; may transfer well to open-ended H5P tasks.
Continue evaluating (pending re-run)	Llama 3.3 70B / Trendyol	Highest absolute Tier 1 scores, but only after a same-item re-evaluation and a hosting-cost review; Trendyol is not confirmed as an improvement over its base.
Deprioritize	Foundation-Sec-8B	Consistent regression relative to the Llama 3.1 base.
Deprioritize	ZySec-7B	Largest consistent loss; topic-level regressions on email, foundations, and regulatory content.
Deprioritize	Encoder-only paths (CyBERTuned)	Useful as a scientific control only; not suitable for H5P generation or explanations.
Conditional	BaronLLM v2	Large delta over a weak Qwen3 base but modest absolute score; worth Tier 2 testing only if the base was misconfigured and structured-output performance is confirmed.
Conditional	Qwen3 14B (base)	Weak Phase 1 MCQ performance; prompt and serving configuration should be investigated before ruling the model out.

Several dimensions remain unranked pending later evaluation phases: valid H5P JSON and didactic quality (Tier 2, the primary Scale-C use case), German-language content (Tier 3), refusal behavior on harmful or exploit-style prompts, and a qualitative error analysis of malware/threat failure modes, which has not yet been conducted. Until Tier 2 is run, the appropriate summary is that this study has narrowed the field on cybersecurity MCQ performance without yet selecting Scale-C’s production model.

The methodological contribution of this study is a pairwise, multi-source, topic-labeled benchmark tied to a real curriculum taxonomy and scored under two independent topic views, demonstrating that cybersecurity fine-tuning evaluations reporting only a single accuracy figure can be misleading, both about which model wins and about whether fine-tuning helped at all. For Scale-C specifically, the practical lesson is to evaluate the same protocol used in production: MCQ, structured JSON, locale, and safety, on items matching the course topics, with frozen manifests and reported n/N.

5.4 Limitations

The following methodological constraints apply to the results reported in this study:

- Reported accuracies come from a single evaluation run per model, with no repeated sampling or multiple random seeds; the deltas in Table 4.4 are therefore point estimates without confidence intervals, and small differences, such as the +0.5 pp CyBERTuned gain, should not be over-interpreted as statistically significant.
- The Llama 3.3 70B base model's NLI evaluation is incomplete: only 327 of the 573 total items had been scored at the time of writing, and only 182 of those were answered correctly; any NLI-based ranking involving this pair should be treated as provisional until the run is completed on the full item set.
- The two topic classifiers were each run once over their respective item pools; beyond the aggregate rank correlation reported in Section 5.1 ($\rho \approx 0.97$), their agreement has not been separately validated against a human-labeled gold-standard topic assignment.
- Only validated, implemented taxonomy labels are presented as classification results; proposed dimensions, risk, difficulty, cognitive skill, and capability tier, are unvalidated and are not used in the reported results.
- Support counts are reported for the topics and models shown in Table 4.6, but not exhaustively for every topic-model combination in the full taxonomy; readers should treat any topic-level figure without a visible n as provisional until the complete per-cell counts are published.
- NLI and embedding classifiers measure related but distinct properties and are treated as complementary views rather than interchangeable metrics.
- A qualitative error analysis of failure cases in malware and advanced-threat topics has not yet been completed; such an analysis would strengthen the study's scientific contribution and is recommended before final submission.

5.5 Conclusion and Future Work

Scale-C is an H5P-based computer security course that may eventually employ a large language model to draft activities, explanations, and localized content. Before committing to a model, this study conducted a benchmarking evaluation of six base/fine-tuned pairs, using public cybersecurity benchmarks mapped to the Scale-C taxonomy, under Tier 1 multiple-choice evaluation with two topic-labeling views (NLI and embedding). The central question, whether cybersecurity fine-tuning reliably outperforms a solid general-purpose base under this protocol, is answered in the negative as a general rule: the outcome depends on the pair, the topic, and the topic-labeling method.

Fine-tuning helps some pairs and harms others: Mistral → Lily (+7 pp) and Qwen3 → Baron (+14–17 pp) improve on both classifiers; Llama 3.1 → Foundation-Sec and Zephyr → ZySec regress; and Llama 3.3 → Trendyol splits by classifier (+9.4 pp NLI, -7.2 pp embedding). A cybersecurity fine-tuned label is not evidence of improved benchmark performance. Absolute leadership also depends on the view: Trendyol leads on NLI (65.1%) and Llama 3.3 leads on embedding (75.7%, subject to denominator caveats), while Lily and Llama 3.1 8B remain credible options at 7B–8B scale for a university budget. Topic-level analysis shows that identity, network, and certificate-style MCQs are relatively strong across models, whereas malware and advanced-threat topics remain difficult

even for 70B systems, so human review should stay in the loop for those modules. Finally, the two topic labelers largely agree ($p \approx 0.97$) but diverge on the overall leader and the 70B fine-tuning verdict, which is reason enough not to lean on a single automatic labeling pipeline for model selection.

Phase 1 covered MCQ performance only. H5P JSON generation, didactic quality, German content, and safety were not tested, so this study does not name a final production model; instead, it provides a shortlist for the next evaluation phase (Table 5.1). For Scale-C, the work contributes a reusable evaluation pipeline (taxonomy, dataset builder, pairwise metrics, and dual topic views) together with an evidence-based shortlist rather than a deployment decision. More broadly, it demonstrates that single-number cybersecurity LLM leaderboards can mislead, and that pairwise comparison combined with topic-level breakdown reveals where fine-tuning helps, where it hurts, and where the field as a whole continues to struggle with hard threat content.

Future work should prioritize three steps: (i) conducting the Tier 2 evaluation of H5P and JSON-schema output with rubric-scored explanations; (ii) completing the 70B re-evaluation on a frozen, shared item set; and (iii) adding safety checks and Tier 3 German-language evaluation where the course requires them. Once these steps are complete, Scale-C will be positioned to commit to a single model, or to a split configuration, for example a smaller model for drafting and a larger model for review, supported by evidence that reflects actual platform use.

Figure References

ID	Figure	Purpose
Figure 1	Benchmark–Framework Association Matrix	Maps benchmarks to cybersecurity frameworks
Figure 2	Pipeline	Shows the end-to-end evaluation workflow
Figure 3	Global accuracy (NLI vs. embedding)	Presents accuracy across the two evaluation views
Figure 4	Encoder control vs. generative models	Isolates the encoder-vs-generative architecture gap
Figure 5	Fine-tuning delta by pair (NLI vs. embedding)	Compares pairwise fine-tuning effects
Figure 6	Base vs. fine-tuned accuracy within each pair	Shows base-vs-fine-tuned comparison logic
Figure 7	Course-by-course performance heatmap	Shows model accuracy by curriculum group
Figure 8	Fine-topic accuracy by model	Supports auditability of topic-level results

Table References

ID	Table
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Table 3.2	Benchmark sources and associated frameworks
Table 3.3	Eight course groups (curriculum-level rollup)
Table 4.1a	Phase 1 evaluation pools (scorable Tier 1)
Table 4.1b	Scorable items by benchmark family
Table 4.1c	Scorable items by course group
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Table 4.3	Fine-tuning outcome categories
Table 4.4	Pairwise fine-tuning deltas (data/eval_1)
Table 4.5	Strongest course groups (embedding view, selected models)
Table 4.6	Difficult fine topics (NLI% / embedding%)
Table 5.1	Recommended status for Tier 2 testing

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